Clinical Features

In any given patient, **polymorphic light eruption (PMLE)** outbreaks typically affect the same sun-exposed sites, usually symmetrically, although the extent may gradually spread or recede over time. Not all exposed skin is involved—lesions often occur on areas that are intermittently exposed or normally covered, while large regions may remain spared. A notable example is **juvenile spring eruption**, which predominantly affects boys in spring and presents with pruritic papules and vesicles localized to the ear helices; typical PMLE may coexist in some cases. Systemic symptoms are uncommon, but rare instances may include malaise, nausea, or other nonspecific sensations.

Cutaneous Lesions

PMLE exhibits a wide range of morphologic presentations, all likely sharing similar pathogenesis and prognosis. Described forms include: - **Papular** - **Papulovesicular** - **Plaque** - **Vesiculobullous** - **Insect bite-like** - **Erythema multiforme-like**

Pruritus alone may rarely be the sole manifestation, though a macular variant has not been clearly documented. The **papular form**—characterized by small or large, discrete or confluent lesions that often cluster—is the most common, followed by papulovesicular and plaque types; the remaining forms are infrequent. An "eczematous" variant has been reported, but such cases likely represent mild **chronic actinic dermatitis** or photo-exacerbated **seborrheic** or **atopic eczema**.

Different morphologic variants can occur simultaneously at various body sites in the same individual. For instance, diffuse facial erythema and swelling may accompany typical papular lesions elsewhere. A variant known as **benign summer light eruption**, recognized in continental Europe, refers to small papular PMLE that spares the face and develops after several days of continued sun exposure.

Laboratory Tests

Histology

Histopathologic findings in PMLE are characteristic but not pathognomonic and vary with lesion morphology. Common features include: - Moderate to dense **perivascular lymphocytic infiltrate** in the upper and mid dermis - Predominance of **T cells**, with occasional **neutrophils** and rare **eosinophils** - **Upper dermal and perivascular edema** - **Endothelial cell swelling**

Epidermal changes, when present, may include: - Variable **spongiosis** - Occasional **dyskeratosis** - **Exocytosis** of lymphocytes - **Basal cell vacuolization**

Blood Tests

Routine laboratory evaluation should include assessment of: - **Anti-nuclear antibodies (ANA)** - **Extractable nuclear antigen (ENA) antibodies**

These tests help exclude **subacute cutaneous lupus erythematosus** or other forms of lupus. If photosensitivity is atypical or there is clinical suspicion, measurement of **red blood cell protoporphyrin** levels is recommended to rule out **erythropoietic protoporphyria**.

Phototesting

- **Monochromator phototesting** confirms photosensitivity in up to 50% of PMLE cases but lacks specificity for distinguishing PMLE from other photodermatoses.
- **Provocation testing** using a solar simulator or broadband light source—sometimes repeated over 2–3 consecutive days—can often reproduce the eruption. This may aid diagnosis, particularly when clinical and historical features are inconclusive, and allows for biopsy confirmation if needed.

Differential Diagnosis

See Box 90-1.

Complications

While most patients experience only recurrent skin eruptions, a small subset may develop **lupus erythematosus**. Additionally, there is an increased prevalence of prior PMLE among individuals diagnosed with lupus, suggesting a possible association between the two conditions.